UMC

LOGIC_MIXED_MODE11-1P8M-MMC _AL_L110AE-CALIBRE-LVS-1.2_P4

Application Notes

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Preface

This application note provides users a brief introduction to switch/option usages, and the environment settings in the command file.

In addition, this application note offers some important messages about the process or device features in the 110nm process node.

Chapter 1. Switch and Option Usage

1. If devices dimensions need to be checked, uncomment the “//” in “// TRACE PROPERTY” statement.

2. Customers should revise the name of "runset.tec" to the correct LPE file name, and turn on PEXRUN while running LPE file.

3. If customers would like to stop running when the tool detects an error in soft-check, please turn on the option: "LVS ABORT ON SOFTCHK YES"

4. If customers would like to SEPGND, please turn on the option: " #DEFINE SEPARATE_GND "

Please refer to 1-1. Switch Information for details.

1-1. Switch Information

(1) LVS Switch Setting

LVS EXECUTE ERC LVS ERC checking
Switch for ERC checking. This switch is YES as default.

(2) LVS DEFINE Setting

Note: Warning!! Do not turn on 2 options at the same time. If do that, a fatal error will occur.

#DEFINE PEXRUN_MC_HSPICE
TO RUN CALIBRE LVS + XRC with Monte Carlo model.
This switch is OFF as default.

#DEFINE PEXRUN
TO RUN CALIBRE LVS+XRC, please turn on the setting for INCLUDE "runset.tec"
This switch is OFF as default.

#DEFINE RCFLOW_CCI
TO RUN CALIBRE LVS+STARRC
This switch is OFF as default.

#DEFINE RCFLOW_QCI
TO RUN CALIBRE LVS+QRC
This switch is OFF as default.

Chapter 2. Environment Setting

To ensure the LVS command file is executed correctly, please make sure to select the correct METAL-OPTION and set the environment of run deck before using this command file. Please refer to the following section for important reminders.

2-1. Layout Environment Set Up

Please set up LAYOUT PATH, LAYOUT PRIMARY, SOURCE PATH and SOURCE PRIMARY according to your design before using CALIBRE LVS.

2-2. Layer Mapping Set Up

Please fill in the layer number information in layer mapping definition. Incorrect layer mapping information will cause serious mistakes.

2-3. Metal option Set Up (Default: TOP_METAL_ME8_2T)

This LVS package includes several run sets separated by different metal structures.
Please refer to the following table to select the necessary run set:

```
#DEFINE TOP_METAL_ME4_1T    // To switch LVS to 1P4M1T
#DEFINE TOP_METAL_ME5_1T    // To switch LVS to 1P5M1T
#DEFINE TOP_METAL_ME5_2T    // To switch LVS to 1P5M2T
#DEFINE TOP_METAL_ME6_1T    // To switch LVS to 1P6M1T
#DEFINE TOP_METAL_ME6_2T    // To switch LVS to 1P6M2T
#DEFINE TOP_METAL_ME7_1T    // To switch LVS to 1P7M1T
#DEFINE TOP_METAL_ME7_2T    // To switch LVS to 1P7M2T
#DEFINE TOP_METAL_ME8_2T    // To switch LVS to 1P8M2T
```

Note: Do not turn on 2 or more metal options at the same time. If do that, a fatal error will occur**.**

2-4 MMC option Set Up (Default: NONE_MMC)

```
#DEFINE NONE_MMC // NO MMC
#DEFINE M23_MMC    // MMC between ME2 & ME3
#DEFINE M34_MMC    // MMC between ME3 & ME4
#DEFINE M45_MMC    // MMC between ME4 & ME5
#DEFINE M56_MMC    // MMC between ME5 & ME6
#DEFINE M67_MMC    // MMC between ME6 & ME7
```

Note:(1)Do not turn on 2 or more MMC options at the same time. If do that, a fatal error will occur.

(2)MMC is metal option dependence, please refer to TLR.

EX: If TOP_METAL_ME6 is selected, then only M65_MMC or M54_MMC will work.

If TOP_METAL_ME5 is selected, then only M54_MMC or M43_MMC will work.

2-5 MMC Options (Default: 1_0FFMIM)

#DEFINE 2_0FFMIM // 2.0fF/um^2 Metal/Metal Capacitor (only RF MIMCAP)
#DEFINE 1_5FFMIM // 1.5fF/um^2 Metal/Metal Capacitor (MM & RF MIMCAP)
#DEFINE 1_0FFMIM // 1.0fF/um^2 Metal/Metal Capacitor (MM & RF MIMCAP)

2-6 Separate ground option (Default: turn off)

#DEFINE SEPARATE_GND To switch LVS to use SEPGND layer

The layer SEPGND is used to separate different ground nets within the same P-substrate. Warning!! If the switch is turned on, the ERC error "SEPARATE_GROUND" will help to highlight the layer SEPGND in layout. Designer should do the layout review and check if this highlight is expected or not.

2-7 IO voltage option setup (Default 3.3V_IO)

#DEFINE 2.5V_IO //SELECT 2.5V IO
#DEFINE 3.3V_HG_IO //SELECT 3.3V HG IO
#DEFINE 3.3V_IO //SELECT 3.3V IO
#DEFINE 5V_LVT //SELECT 5V LVT IO
#DEFINE 5V_RVT //SELECT 5V LVT IO

Note: Do not turn on 2 or more metal options at the same time. If do that, a fatal error will occur**.**

2-8 TOP/Thick Metal option setup

#DEFINE Thick_Metal_12K // TOP-Metal sheet resistance (W= 0.72 um) Typ.= 23 mOhm/sq
#DEFINE Thick_Metal_20K // TOP-Metal sheet resistance (W= 1.2 um) Typ.= 15 mOhm/sq

#DEFINE Thick_Metal_40K   // TOP-Metal sheet resistance (W= 2.06 um) Typ.= 7 mOhm/sq

Note: Do not turn on 2 or more metal options at the same time. If do that, a fatal error will occur**.**

2-9 Process Option setup (Default: HS_SP_LL)

```
#DEFINE HS          // DEVICE OF HS PROCESS
#DEFINE SP          // DEVICE OF SP PROCESS
#DEFINE LL          // DEVICE OF LL PROCESS
#DEFINE HS_SP       // DEVICE OF HS_SP PROCESS
#DEFINE HS_LL       // DEVICE OF HS_LL PROCESS
#DEFINE SP_LL       // DEVICE OF SP_LL PROCESS
#DEFINE HS_SP_LL    // DEVICE OF HS_SP_LL PROCESS
```

Note: Do not turn on 2 or more metal options at the same time. If do that, a fatal error will occur**.**

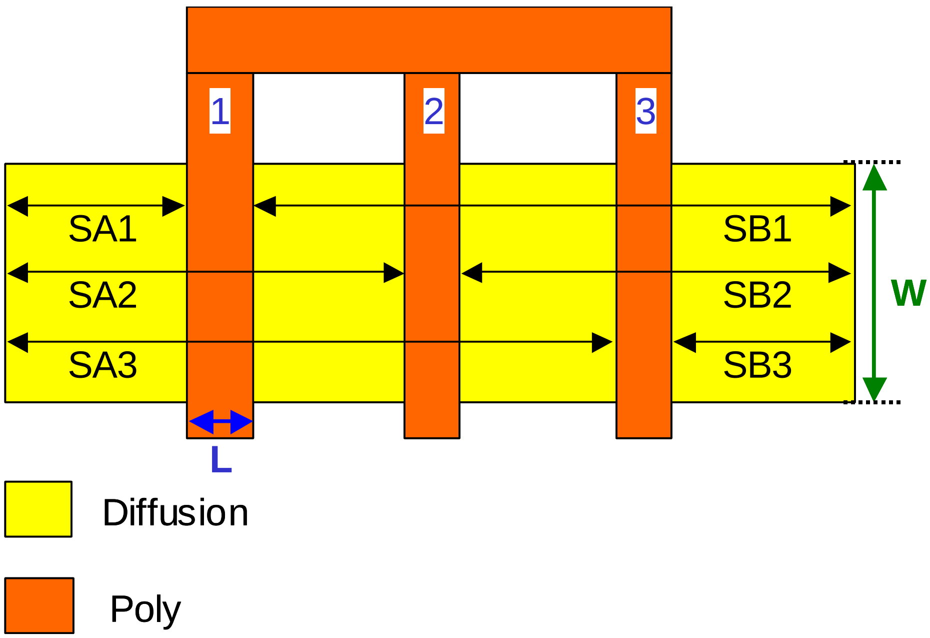
2-10 Other option setup

#DEFINE BACK_TO_BACK_DIODE // Default is turn off --> don't recognize back to back diode
#DEFINE CHECK_DIFF_Used_As_Conductor
#DEFINE METAL_INCLUDE_DUMMY_LAYER   // To switch of Metal layer include Metal Dummy layer.

Chapter 3. Process Effects Supported in LVS

3-1. LOD (Length of Diffusion)

STI process causes the compressive stress in the channel. The stress will impact carrier mobility and exhibit a significant sensitivity to layout.



Extraction of Instance Parameters

SA: distance between OD edge to poly from one side
SB: distance between OD edge to poly from the other side
SD: poly space for multiple finger device (Note*)

Note: Please be noted that the MOSFETs with multi-fingers will be flattened for one-finger transistors at LPE stage. The SD will not be output in the extracted net-list.

Chapter 4. CCI Flow (Star-RC)

4-1 XRC Flow

A. Files Preparation

i. Calibre LVS file:
   Select the needed metal option, open the switch (PEXRUN) and include LPE file(.tec).

ii. XRC LPE file (.tec)

iii. XRC mapping file (bundled in LVS package)

B. Steps for XRC Flow

Step 1 : Source license of Calibre and XRC.

Step 2 : Un-tar the xRC LPE package and Calibre LVS package.

Step 3 : Turn on the switch

i. Choose the right metal option.

ii. Turn on the switch as below.
   #DEFINE PEXRUN.

iii. Setup layout/source path and topcell name in LVS file.

iv. Include the tec file in LVS file. (INCLUDE “runset.tec”)

Step 4 : Execute LVS compare by following command.

> calibre –lvs –hier <Calibre LVS file>

Step 5 : Perform the following commands to generate parasitic netlists.

> calibre –xrc –pdb –c <Calibre LVS file>

> calibre –xrc –fmt –c <Calibre LVS file>

4-2 CCI Flow

Star-RC Overview

Star-RC is a next-generation overdrive option to the Star-RC™ layout parasitic extraction tool that extracts connected databases.

Star-RC can be used at any time during the physical design cycle to extract accurate parasitics.

- Star-RC reads Milkyway™ Place & Route, LEF/DEF, or Hercules connected databases directly, without external processing.
- Extracted parasitics can be written into the Avant! centralized Milkyway database for use by any of the Apollo™ family of analysis and optimization tools.

Since Star-RC gracefully handles designs with LVS violations, including opens and shorts, timing convergence can be ensured before beginning the physical verification cycle.

- Star-RC includes tools for the extraction, verification, and analysis of a design's clock networks.
- Extraction/analysis can be specified within the Star-RC SingleShot™ extraction flow so that the entire tape-out verification extraction can be executed in one step.

A. Files Preparation

i. Calibre LVS file:
   Select <u>the needed metal option</u> and open <u>the switch (RCFLOW_CCI)</u>
ii. StarRC LPE file (.nxtgrd)
iii. CCI mapping file (bundled in LVS package)
iv. CCI_star_cmd (bundled in LVS packages)
v. query.txt (bundled in LVS packages)

B. Steps for CCI Flow

Step 1 : Source license of Calibre and StarRC.
Step 2 : Un-tar the StarRC LPE package and Calibre LVS package.
Step 3 : Turn on the switch
    Choose the right metal option.
    Turn on the switch as below.
    #DEFINE RCFLOW_CCI.
    Setup layout/source path and topcell name
Step 4 : Execute LVS compare by following command.
    > calibre –lvs –hier <Calibre LVS file>
Step 5 : Make a new directory -CCI, then invoke the following command to create a lot of files in the directory - CCI.
    > calibre -query svdb < query.txt | tee query.log
Step 6 : Put corresponding CCI layer mapping file and StarRC nxtgrd file on working directory.
Step 7 : Customize the setting in CCI_star_cmd, then invoke the following command to perform RC extraction to generate parasitic netlist.
    > StarXtract -clean CCI_star_cmd

****NOTE : UMC strongly recommend using both “CCI_star_cmd” and “query.txt” in the LVS package for CCI flow.**

4-3 QCI Flow

A. Files Preparation

i. Calibre LVS file:
   Select the needed metal option and open the switch (RCFLOW_QCI)
ii. QRC LPE files
iii. QCI mapping file (bundled in LVS package)
iv. QCI_qrc_cmd (bundled in LVS packages)
v. query.txt (bundled in LVS packages)
vi. techgen_scr (bundled in LVS packages)

B. Steps for QCI Flow

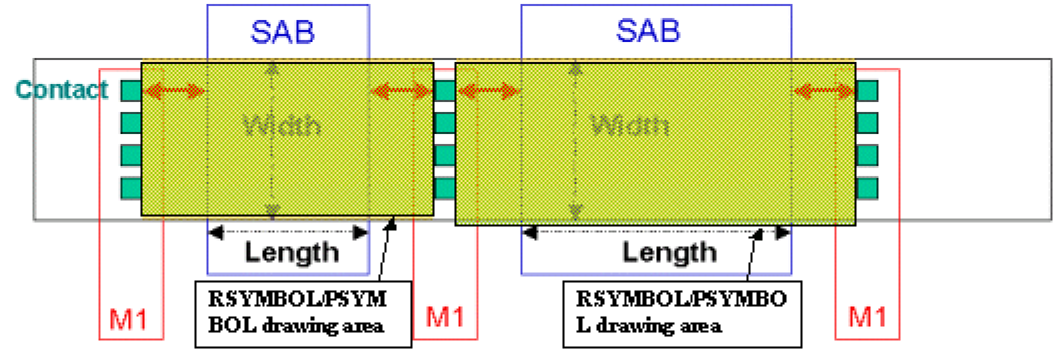
Step 1 : Source license of Calibre and QRC.
Step 2 : Un-tar the QRC LPE package and Calibre LVS package in different directories.
Step 3 : Turn on the switch
    Choose the right metal option.
    Turn on the switch as below.
    #DEFINE RCFLOW_QCI.
    Setup layout/source path and topcell name
Step 4 : Make a new directory - query_output ,invoke the following command to create QCI database in the directory - query_output.
    > calibre -query svdb < query_QCI.txt | tee query.log
Step 5 : Create LVS rundeck "lvsfile" by following command.
    > calibre -E new_calibre_LVSfile calibre_LVSfile
    > calibre2qrc -keep_layer_name -r new_calibre_LVSfile:lvsfile
Step 6 : Get the corresponding layer_setup file from ./qcimap which is in LVS package.
Step 7 : Before performing QRC extraction flow, please use the script, techgen_scr, to do compilation in the directory where QRC LPE package is.

Users would see two new files, RCXspiceINIT and RCXdspfINIT upon completion of compilation.
If the options are shown in the line beginning with CAPGENOPTS in the bottom of the file, RCXspiceINIT or RCXdspfINIT, do not meet your need cause by customized devices in LVS rundeck, users have to run the script to do compilation again after adjusting options.

Step 8 : Invoke the following command to do RC extraction.
    > qrc -cmd QCI_qrc_cmd
    The template of qrc command file - QCI_qrc_cmd can be found in ./lpecmd in the LVS package.

**NOTE : UMC strongly recommend using both "QCI_qrc_cmd", "query.txt" and "techgen_scr" in the LVS package for QCI flow.

Chapter 5. Non-Salicide resistor symbol drawing



The RSYMBOL/PSYMBOL drawing area outside of SAB is follow TLR SAB to CONT spacing

Note: Please refer device information to draw resistor

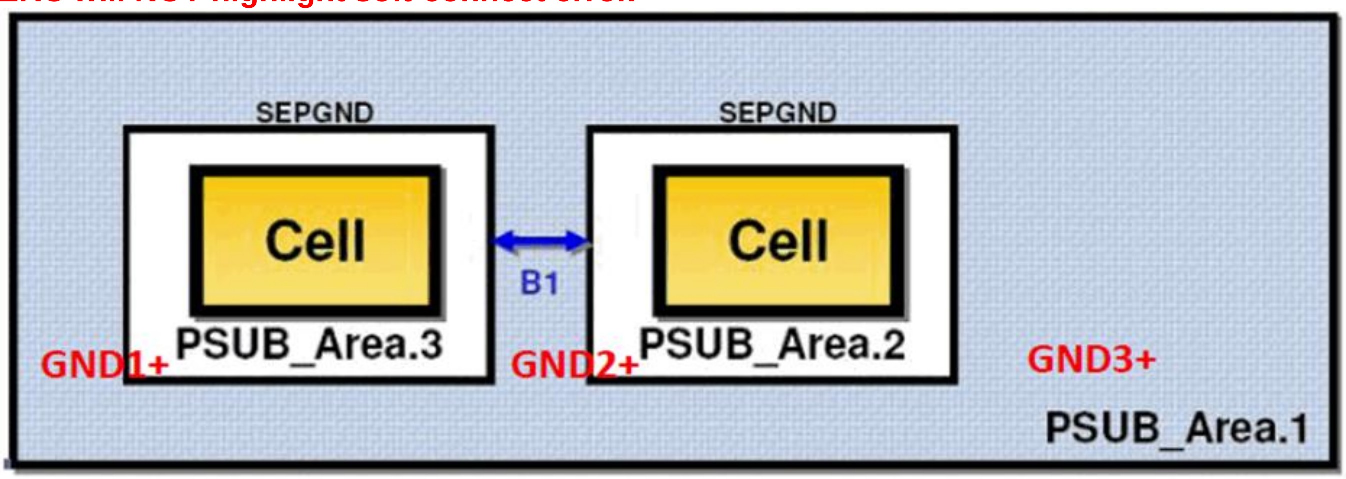
Chapter 6. SEPGND Layer Usage Guidance

CAD Verification Layer

SEPGND						
Tech Layer Name	Tech Purpose Name	GDS No.	Data Type	Tech Layer No.	Tech Purpose No.	Description
SEPGND	drawing	99	0	99	252	Separate Ground Marker

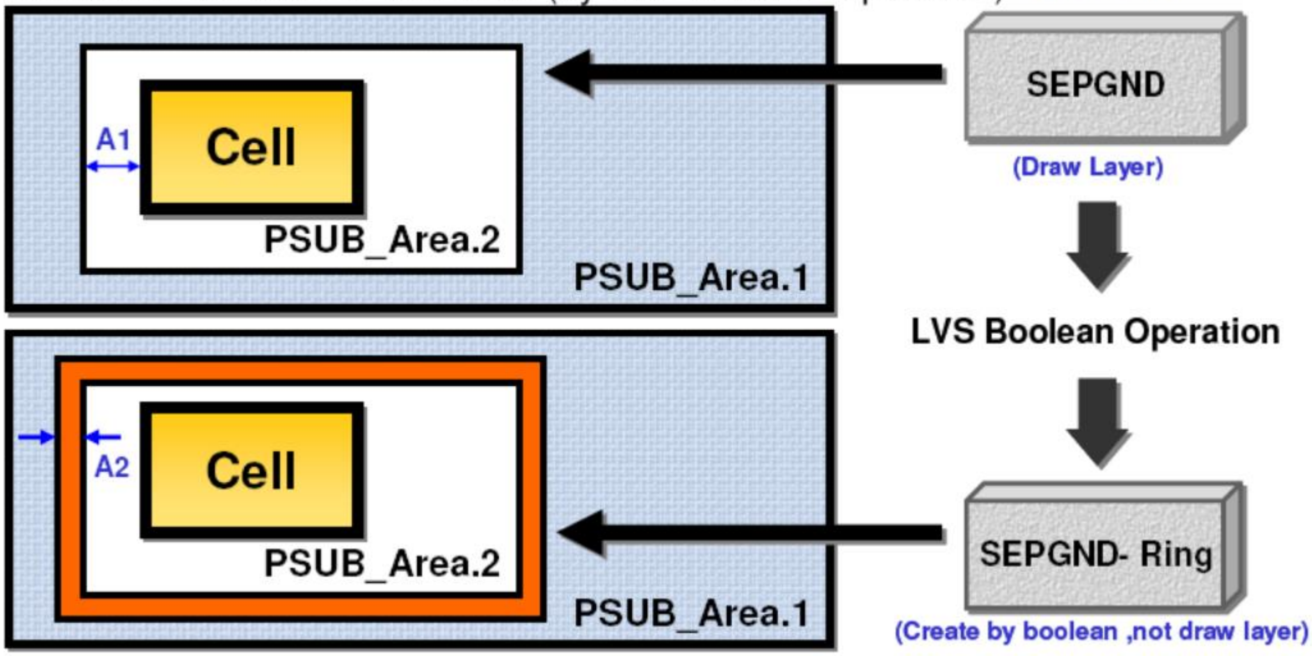
1. **The SEPGND-Ring is used to separate different GND nets within the same P-Well.**
2. **The purpose is to meet analog design and customer specific needs.**
3. **The P-sub inside SEPGND layer will be completely isolated by connectivity from the chip substrate in both LVS and ERC/soft-connect verification. However, physically, the P-sub inside the SEPGND layer is still the silicon P-substrate and has the same connectivity. Therefore, all P-subs inside SEPGND are connected to silicon P-substrate.**

For example: There are three potentials in layout. If the SEPGND is used as below, the ERC will NOT highlight soft-connect error.

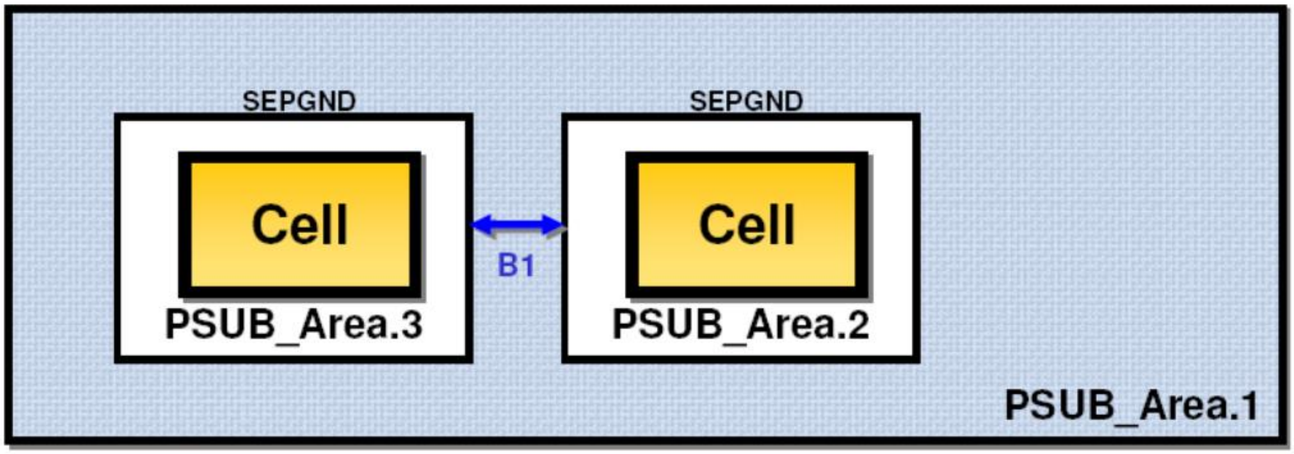


4. **We do NOT recommend customers to use SEPGND layer to isolate P-sub regions. It is better to use NWELL ring combined with DNW to isolate P-sub regions.**
5. **Designers should do layout review with checking on ERC violations and warnings related to the connections of SEPGND layer for proper usage of SEPGND layer.**
6. **For usage, please follow the following rules:**

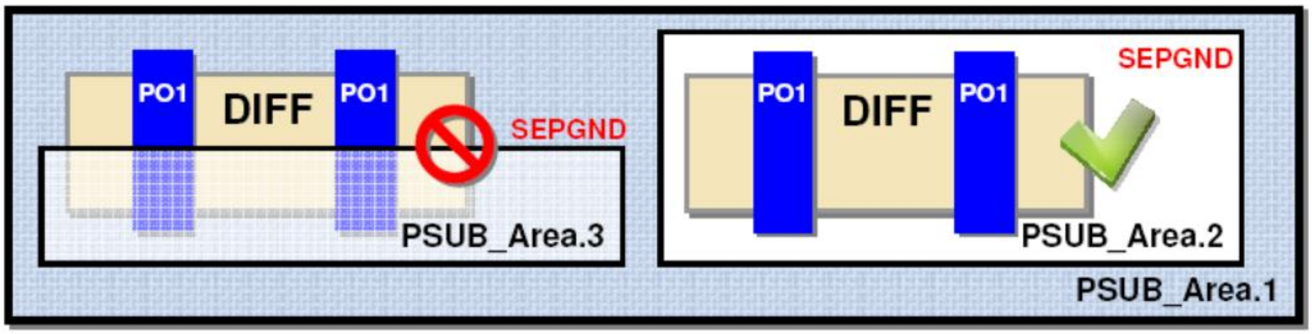
The SEPGND-Ring used to separate different potential (PSUB)
A1. SEPGND enclosure of Cell >= 0.001 um
A2. Exact extension of SEPGND (by LVS Boolean Operation) 0.001 um



B1. Between two SEPGND , the better drawing spacing of them is >= 0.003um



C. Please be noted that SEPGND layer MUST enclose the entire DIFF region



D1. SEPGND layer MUST be separate from well layers by >= 0.002um and can't overlap with well layers